

Attention Mechanisms in Neural Networks

The attention mechanism was introduced in "Attention Is All You Need" by Vaswani et al. (2017). It allows models to focus on relevant parts of the input when generating output.

Key concepts:

- Query, Key, Value (QKV) triplets
- Scaled dot-product attention
- Multi-head attention for parallel processing

The transformer architecture has become the foundation for modern NLP models including BERT, GPT, and T5.

Retrieval-Augmented Generation (RAG)

RAG combines retrieval systems with generative models. First introduced by Lewis et al. (2020), RAG addresses the limitation of LLMs being limited to their training data.

Components of a RAG system:

1. Document store with vector embeddings
2. Retriever to find relevant documents
3. Generator (LLM) to produce responses

Benefits include reduced hallucination, up-to-date information, and source attribution.

LangChain and LangGraph Framework Overview

LangChain is an open-source framework for building LLM applications. Key features include modular components, integration with 50+ LLM providers, and built-in RAG utilities.

LangGraph extends LangChain for stateful applications with graph-based state management, support for cycles and loops, and human-in-the-loop workflows.